Biochemistry Cumulative Examination
October 4, 2002

1. Carbohydrates are very important biological molecules. Give the general molecular formula for this class of molecules. 5 points

2. Give a brief written definition of what a sugar is (in chemical terms). 10 points

3. What is the difference between a ketose and an aldose? 10 points

4. A pyranose and a furanose? 10 points

5. A hexose and a pentose? 10 points

6. Draw glucose in Fischer projection, being sure to show stereochemistry, and then in cyclic form, being sure to show the normal conformation. On this latter drawing, clearly indicate the anomeric carbon. 20 points

7. Draw ribose, being sure to show stereochemistry. Clearly indicate the hydroxyl that is missing in deoxyribose. 10 points

8. Draw any sugar attached to ethanol by a glycosidic bond. 15 points

9. Draw a short section (3 monomers) of starch and of cellulose, clearly showing the difference(s) between the two polysaccharides. 10 points